

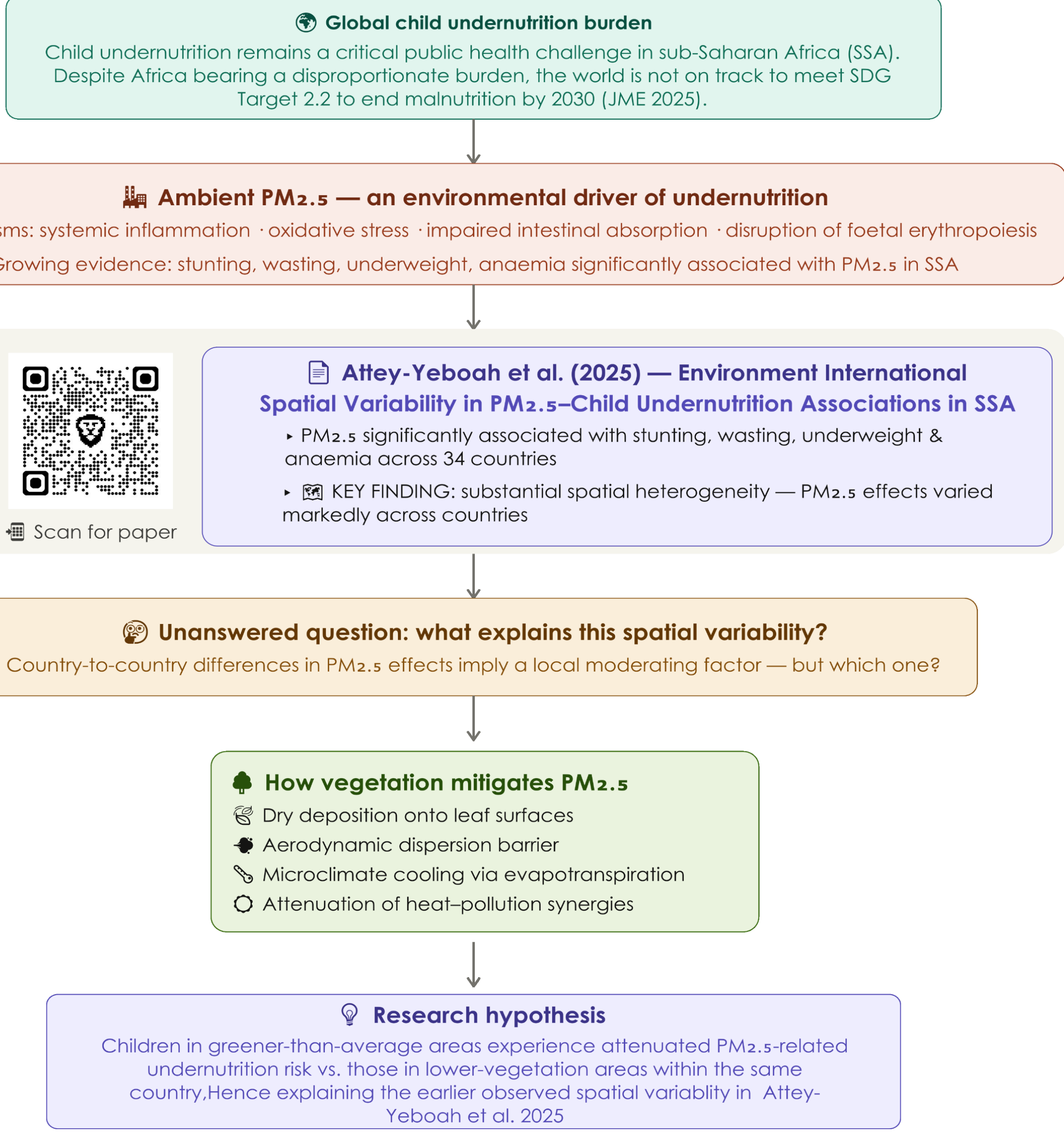
Spatial Variability in Ambient PM_{2.5} Exposure Effects on Child Undernutrition in Sub-Saharan Africa: The Moderating Role of Green Spaces

Patrick Attey-Yeboah¹, Paul K. Adjorlolo², Justice Moses K. Aheto², John Molitor³, Eric S. Coker⁴, Adeladza Kofi Amegah^{1,5*}

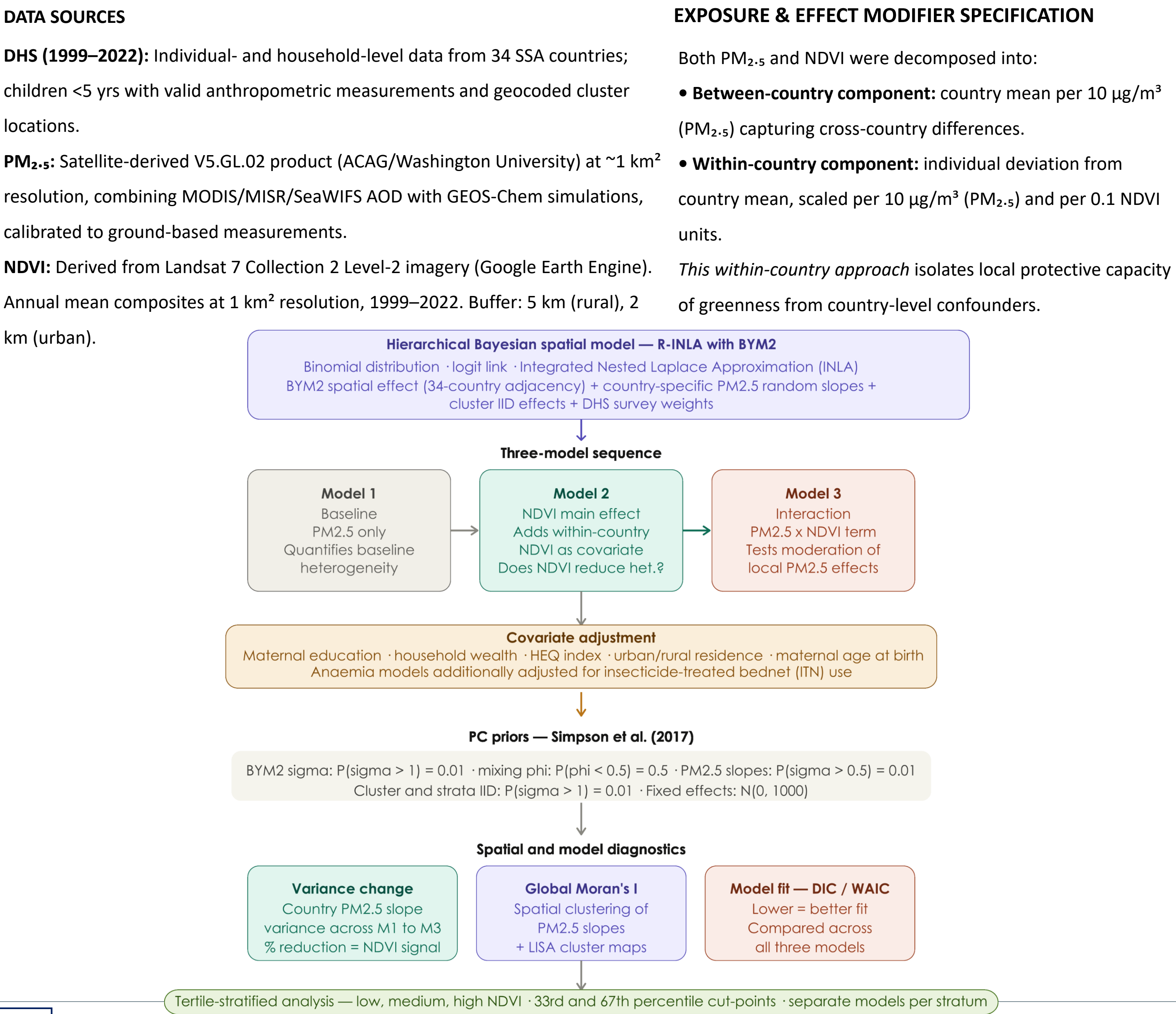
¹ University of Cape Coast, Ghana | ² University of Ghana | ³ Oregon State University, USA | ⁴ Environmental & Global Health, LLC, Canada | ⁵ School of Public Health, UCC



BACKGROUND



RESEARCH APPROACH



FINDINGS

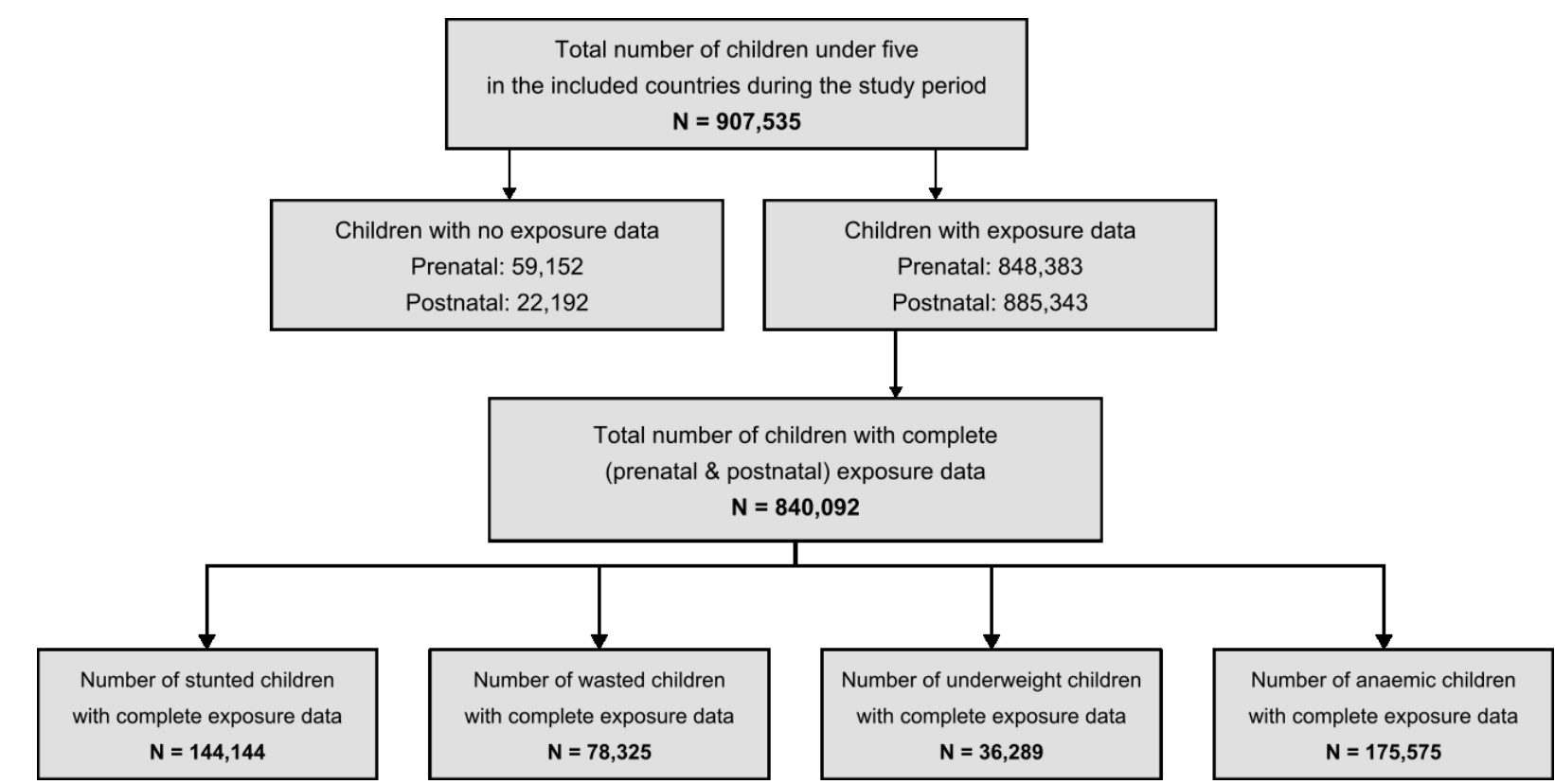
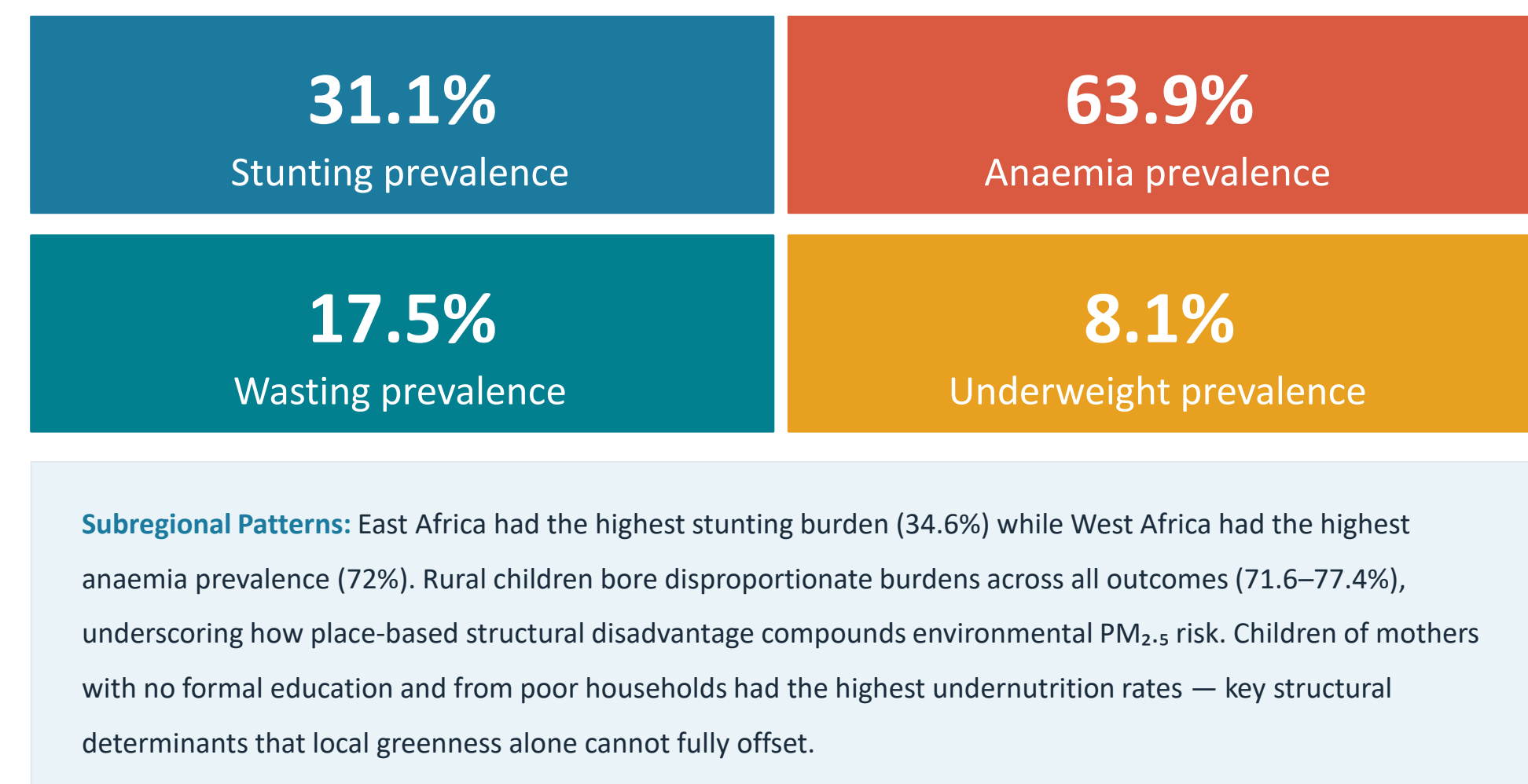


Figure 1. Flowchart of study sample selection process.



VARIANCE CHANGE

Country-specific PM_{2.5} slope posterior variance across models.

Outcome	Exp.	M1 Var. (95% CrI)	ΔM1→M2	ΔM1→M3
Stunting	Pre	0.078 (0.047–0.137)	–0.9%	–0.8%
Stunting	Post	0.048 (0.030–0.083)	+1.9%	+2.3%
Wasting	Pre	0.033 (0.018–0.064)	+1.5%	+0.6%
Wasting	Post	0.023 (0.012–0.050)	0.0%	+1.3%
Underweight	Pre	0.056 (0.031–0.114)	–0.5%	–0.9%
Underweight	Post	0.090 (0.052–0.175)	+1.9%	+2.0%
Anaemia	Pre	0.107 (0.065–0.197)	+4.1%	+4.7%*
Anaemia	Post	0.154 (0.092–0.283)	–0.6%	–0.6%

NDVI explains ≤4.7% of between-country PM_{2.5} effect heterogeneity. M1=Baseline; M2=NDVI Main; M3=Interaction.

GLOBAL MORAN'S I

Spatial clustering of country-specific PM_{2.5} slopes across models.

Outcome	Pre Baseline	Pre NDVI Main	Pre Interaction	Post Baseline	Post Interaction
Stunting	–0.119 (p=0.747)	–0.128 (p=0.768)	–0.128 (p=0.768)	0.128 (p=0.125)	0.125 (p=0.130)
Wasting	0.181 (p=0.060)†	0.173 (p=0.069)†	0.161 (p=0.080)†	0.171 (p=0.075)†	0.130 (p=0.126)
Underweight	–0.219 (p=0.915)	–0.218 (p=0.914)	–0.221 (p=0.918)	–0.087 (p=0.659)	–0.098 (p=0.687)
Anaemia	–0.020 (p=0.454)	–0.020 (p=0.453)	–0.014 (p=0.438)	–0.225 (p=0.905)	–0.217 (p=0.895)

* Borderline clustering (p<0.10). Wasting showed progressive attenuation with NDVI inclusion — the best spatial evidence of greenness-mediated effect moderation.

Acknowledgement

The study was funded by National Institute of Environmental Health Sciences of the National Institutes of Health under Award Number U01ES036147. The content of this research is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health

- Amegah, A. K., Quansah, R., & Jaakkola, J. J. K. (2014). Household air pollution from solid fuel use and risk of adverse pregnancy outcomes: A systematic review and meta-analysis of the empirical evidence. *PLOS ONE*, 9(12), e113920. <https://doi.org/10.1371/journal.pone.0113920>
- Attey-Yeboah, P., Adjorlolo, P. K., Aheto, J. M. K., Molitor, J., Coker, E. S., & Amegah, A. K. (2025). Spatial variability in ambient PM_{2.5} exposure effects on child undernutrition in Sub-Saharan Africa: The moderating role of green spaces. *Environmental Research*.
- Barker, D. J. P. (1995). Fetal origins of coronary heart disease. *BMJ*, 311(6998), 171–174. <https://doi.org/10.1136/bmj.311.6998.171> ICF International. (2022). *Demographic and Health Surveys Program*. <https://dhsprogram.com>

PM_{2.5} × NDVI INTERACTION EFFECTS

Outcome	Exposure Period	Between-Country PM _{2.5}	Within-Country PM _{2.5}	NDVI Main Effect	PM _{2.5} × NDVI Interaction
Stunting	Prenatal	0.97 (0.88–1.07)	1.17 (1.14–1.20)	0.99 (0.98–0.99)	1.00 (0.99–1.01)
Stunting	Postnatal	0.98 (0.88–1.08)	1.13 (1.09–1.16)	0.99 (0.98–1.01)	0.99 (0.98–1.01)
Wasting	Prenatal	1.12 (1.02–1.24)	1.09 (1.06–1.12)	0.99 (0.97–0.99)	0.98 (0.97–0.99)*
Wasting	Postnatal	1.13 (1.02–1.25)	1.09 (1.06–1.13)	0.98 (0.97–0.99)	0.98 (0.97–0.99)*
Underweight	Prenatal	1.16 (1.03–1.31)	0.99 (0.95–1.03)	0.99 (0.98–1.02)	0.98 (0.96–0.99)*
Underweight	Postnatal	1.16 (1.03–1.32)	0.95 (0.90–0.99)	0.98 (0.96–1.00)	0.98 (0.96–0.99)*
Anaemia	Prenatal	1.18 (1.04–1.33)	1.01 (0.96–1.06)	0.97 (0.96–0.99)	0.96 (0.95–0.98)*
Anaemia	Postnatal	1.17 (1.02–1.33)	0.99 (0.94–1.06)	1.02 (1.00–1.03)	0.99 (0.98–1.01)

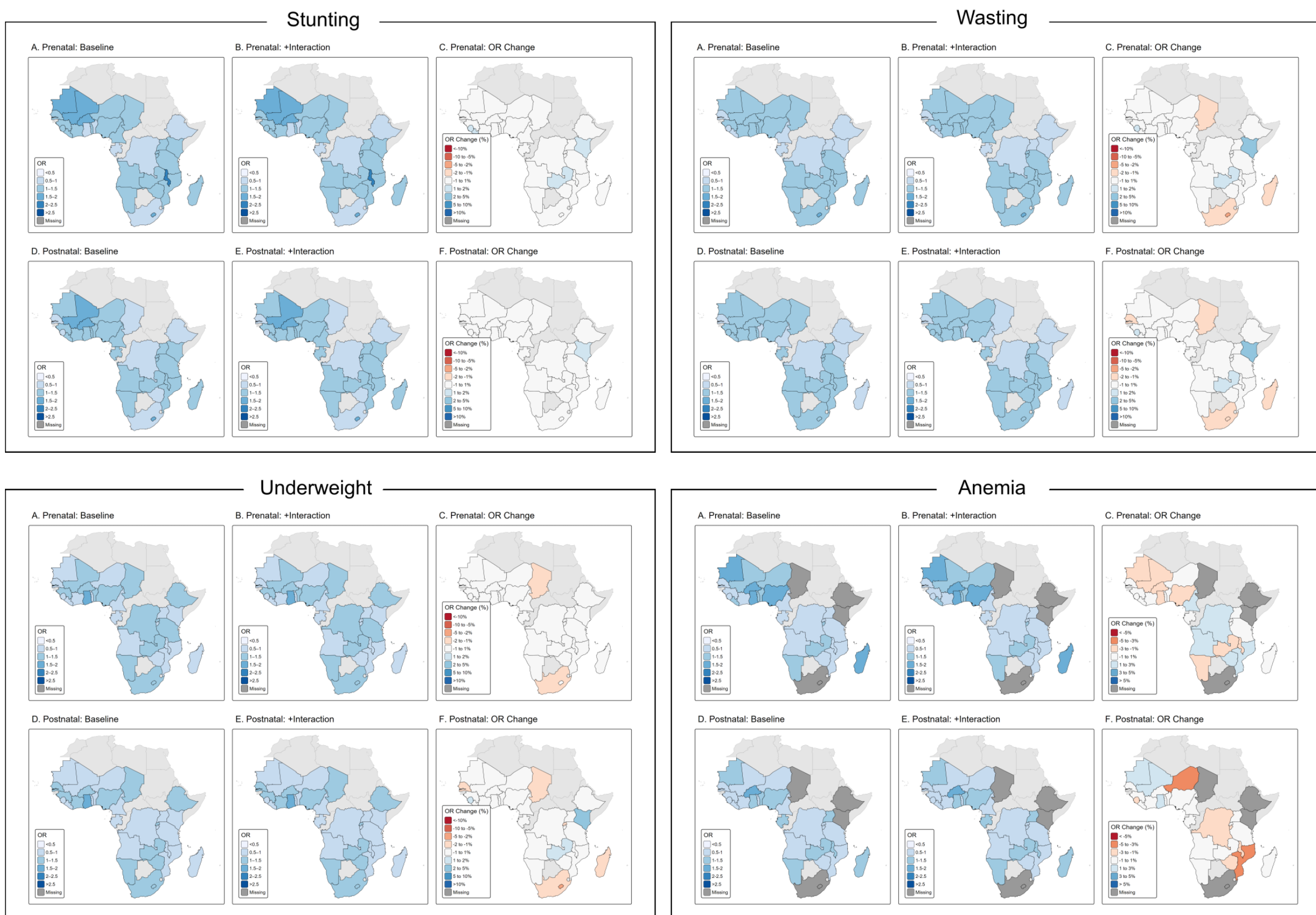
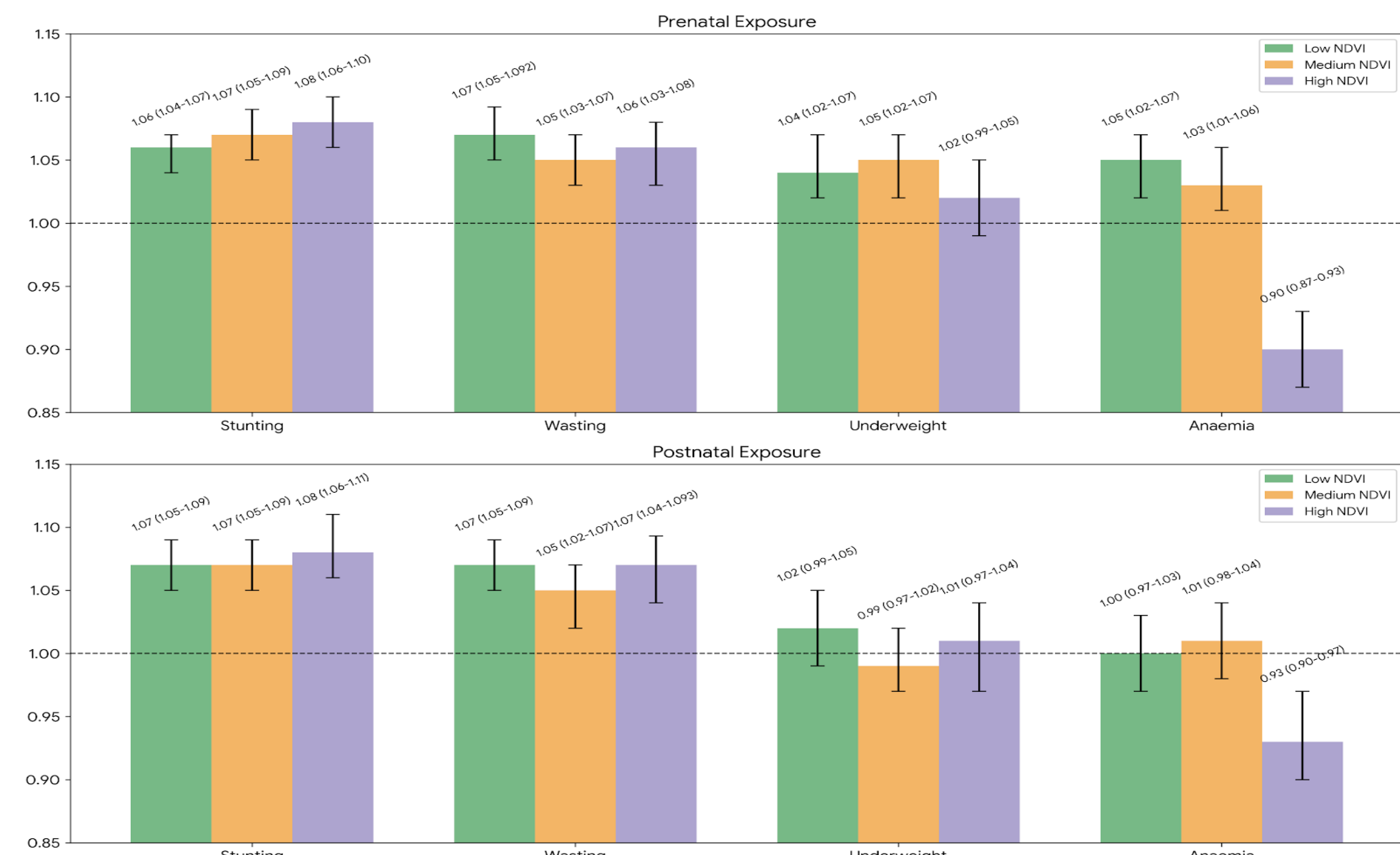


Figure 2: NDVI Modification of Spatial Variability in the PM_{2.5}–Child Undernutrition Association in SSA

STRATIFIED ANALYSIS BY NDVI TERTILE



What We Found

- First continent-wide evidence that local greenness modifies PM_{2.5}–child undernutrition associations in 34 SSA countries.
- Greenness significantly attenuated PM_{2.5}-attributable risk of wasting and anaemia, particularly in the prenatal window.
- Stunting showed no effect modification; underweight showed mixed patterns.
- Anaemia exhibited the strongest modification — PM_{2.5} reversed from harmful to protective in high-NDVI environments.
- Local NDVI explained ≤4.7% of between-country PM_{2.5} effect heterogeneity.

Policy Implications

- Green infrastructure investments (urban greening, tree planting, agricultural intensification) may serve as co-benefit strategies simultaneously improving air quality and reducing child undernutrition attributable to air pollution.
- Limited between-country explanatory power of NDVI underscores that local environmental interventions alone are insufficient. Addressing healthcare access, food security, poverty, and clean household energy are also essential.

Future Research

- Seasonally resolved NDVI metrics are needed — annual means may mask critical dry-season windows when vegetation-mediated PM_{2.5} deposition is minimal.
- Vegetation-type disaggregation (trees vs. shrubs vs. crops) to better characterise PM removal efficiency.
- Longitudinal cohort designs to disentangle greenness timing from cumulative exposure pathways.
- Quasi-experimental studies evaluating greenspace interventions on child health outcomes.

Corresponding Author

Prof. A.K Amegah (P.I),

University of Cape Coast, Ghana

aamegah@ucc.edu.gh

THE 6th
DS-I AFRICA
CONSORTIUM MEETING
16TH - 22ND MAY 2026